

TASK 2 – DATA CLEANING (POWER QUERY)

- Detect and remove duplicates or missing values.
- Standardize column names and data types.
- Perform necessary transformations (split, merge, replace values).
-  Task: Prepare a clean dataset for modelling.

Order Id cleaned

Queries [5]		Table Editor (*Sorted Rows, 1)										Query settings	
												PROPERTIES	
												Name	
												cleaned Orders (2)	
												All Properties	
												APPLIED STEPS	
												Source	
												Navigation	
												Promoted Headers Rename	
												Changed Type default	
												Removed Duplicates in Sorted Rows	
												X Removed Top Rows	
												Trimmed Text	
												split on order date	
												Changed Type1	
												Split on Ship Date	
												Changed Type2	
												Removed split column	
												Filtered Rows	
												Replaced Value	
												Replaced Value1	
												Trimmed Text1	
												Capitalized Each Word	
												Filled Down in country null	
												Delimited # in Region column	
												Changed Type3	
												Removed Columns	
												Capitalized Each Word1 in ...	

1.2 cleaned Data.Discount	1.2 cleaned Data.Profit	A3 cleaned Data.Order Priority
0	2.76	High
0	20.88	Medium
0	6.88	Medium
0.5	-26.055	High
0	106.14	Medium
0.1	36.036	Medium
0	71.46	High
0	746.4078	High
0	746.4078	High
0	3.12	Medium
0.45	-122.8005	High
0	50.13	Medium
0.4	-319.464	Low
0	17.472	Medium
0	17.472	Medium
0.47	-45.9018	Medium
0.1	110.412	Critical
0	7.77	High
0	42.9	Medium
0	253.32	High
0	290.43	Medium
0	26.76	Medium
0.6	-107.856	Medium
0	609.84	Medium
0	45	High
0.5	-21.9	Medium
0	3.6	Medium
0.2	5.5512	High
0	13.68	Medium
0.17	3.3924	Medium
0	22.92	Low
0	20.52	Medium
0	4.65	Medium
0.2	-64.7748	High
0	35.64	High
0.17	6.0084	High

= Table.ReplaceValue("#Replaced Value8","#","",Replacer.ReplaceText,{"cleaned Data.Order Priority"})									
Date	A ₀ cleaned Data.Customer Name	A ₀ cleaned Data.Country	A ₀ cleaned Data.Region.1	A ₀ cleaned Data.Category	A ₀ cleaned Data.Sub-Category	1.2 cleaned Data.Sales	1.2 cleaned Data.Quantity	1.2 cleaned D	
1	Brendan Sued	Benin	Western Africa	Office Supplies	Binders	21.55	1		
2	Arthur Pritchep	United Kingdom	Western Europe	Office Supplies	Paper	139.77	3		
3	Arthur Wiediger	Cuba	Caribbean	Furniture	Labels	38.26	1		
4	Eugene Moren	Sweden	Northern Europe	Office Supplies	Storage	44.865	3		
5	Toby Braumhardt	Algeria	North Africa	Office Supplies	Storage	408.3	2		
6	Joseph Holt	Australia	Oceania	Office Supplies	Binders	120.366	0		
7	Harry Marie	Lebanon	Western Asia	Technology	Binders	174.33	1		
8	Maria Etezadi	United States	Southern Us	Furniture	Storage	2573.82	9		
9	Maria Etezadi	United States	Southern Us	Furniture	Storage	2573.82	9		
10	Magdelene Morse	Canada	Canada	Technology	Binders	314.22	1		
11	Larry Blacks	philippines	Southeastern Asia	Office Supplies	Bookcases	337.6395	3		
12	Dennis Pardue	Malaysia	Southeastern Asia	Technology	Chairs	192.87	1		
13	Ken Lonsdale	New Zealand	Oceania	Technology	Bookcases	912.456	0		
14	Kristina Nunn	United States	Eastern Us	Office Supplies	Labels	36.4	5		
15	Kristina Nunn	United States	Eastern Us	Office Supplies	Labels	36.4	5		

= Table.ReplaceValue("#Replaced Value8","#","",Replacer.ReplaceText,{"cleaned Data.Order Priority"})									
Order ID	A ₀ Order Date	A ₀ Ship Date	A ₀ New Order date	A ₀ New Ship Date	A ₀ cleaned Data.Customer Name	A ₀ cleaned Data.Country	A ₀ cleaned Data.Region.1	A ₀ cleaned Data	
1	BM-2012-85159015-41025	null	05-01-2012	05-01-2012	Brendan Sued	Benin	Western Africa	Office Suppl	
2	ES-2012-AP1091545-40912	null	08-01-2012	08-01-2012	Arthur Pritchep	United Kingdom	Western Europe	Office Suppl	
3	MX-2012-AW1093031-40912	null	08-01-2012	08-01-2012	Arthur Wiediger	Cuba	Caribbean	Furniture	
4	IT-2012-EM14140124-40909	01-01-2012	05-01-2012	05-01-2012	Eugene Moren	Sweden	Northern Europe	Office Suppl	
5	AG-2012-TB112803-40909	01-01-2012	06-01-2012	06-01-2012	Toby Braumhardt	Algeria	North Africa	Office Suppl	
6	IN-2012-H159857-40909	01-01-2012	01-01-2012	01-01-2012	Joseph Holt	Australia	Oceania	Office Suppl	
7	LE-2012-HM486071-40915	null	11-01-2012	11-01-2012	Harry Marie	Lebanon	Western Asia	Technology	
8	CA-2012-ME17320140-40915	null	02-01-2012	11-01-2012	Maria Etezadi	United States	Southern Us	Furniture	
9	CA-2012-ME17320140-40915	07-01-2012	11-01-2012	11-01-2012	Maria Etezadi	United States	Southern Us	Furniture	
10	CA-2012-MM726023-40910	02-01-2012	06-01-2012	02-01-2012	Magdelene Morse	Canada	Canada	Technology	
11	IN-2012-LB16753102-40911	03-01-2012	07-01-2012	07-01-2012	Larry Blacks	philippines	Southeastern Asia	Office Suppl	
12	IN-2012-OP1339078-40911	03-01-2012	09-01-2012	09-01-2012	Dennis Pardue	Malaysia	Southeastern Asia	Technology	
13	ID-2012-KL1664592-40911	03-01-2012	09-01-2012	09-01-2012	Ken Lonsdale	New Zealand	Oceania	Technology	
14	CA-2012-KN16705140-40969	03-01-2012	07-09-2012	07-09-2012	Kristina Nunn	United States	Eastern Us	Office Suppl	
15	CA-2012-KN16705140-40969	01-09-2012	07-09-2012	07-09-2012	Kristina Nunn	United States	Eastern Us	Office Suppl	
16	ID-2012-CM12735130-40911	03-01-2012	08-01-2012	08-01-2012	Chris Mofae	Thailand	Southeastern Asia	Office Suppl	
17	IN-2012-KN164507-40911	03-01-2012	03-01-2012	03-01-2012	Keen Nguyen	Australia	Oceania	Office Suppl	
18	IR-2012-NG83560-40911	03-01-2012	08-01-2012	08-01-2012	Nat Gilpin	Iran	Southern Asia	Office Suppl	
19	IR-2012-JO528060-40911	03-01-2012	07-01-2012	07-01-2012	Unknown	Iran	Southern Asia	Office Suppl	
20	IZ-2012-LW699061-40911	03-01-2012	null	03-01-2012	Unknown	Iraq	Western Asia	Furniture	
21	ES-2012-DO13570139-40911	03-01-2012	07-01-2012	07-01-2012	Dorothy Dickinson	United Kingdom	Northern Europe	Furniture	

Query Settings			
PROPERTIES			
Name			
orders			
All Properties			
APPLIED STEPS			
Removed Columns			
Renamed Columns			
Split Column by Delimiter1			
Removed Columns1			
Renamed Columns1			
Replaced Value2			
Replaced Value			
Replaced Value1			
Removed Top Rows			
Changed Type with Locale			
Replaced Errors			
Removed Top Rows1			
Changed Type1			
Replaced Value3			
Replaced Value4			
Replaced Value5			
Replaced Value6			
Replaced Value7			
Replaced Value8			
Replaced Value9			
Replaced Value10			
Replaced Value11			
Replaced Value12			
Replaced Value13			
Replaced Value14			
Replaced Value15			
Replaced Value16			
Replaced Value17			
Replaced Value18			
Removed Top Rows3			
Changed Type2			
Added Custom			
Added Custom1			
Sorted Rows1			

A ^B C	Order Priority	New Order date	New Ship Date
2.76	High	27-12-2011	05-01-2012
20.88	Medium	30-12-2011	08-01-2012
6.88	Medium	30-12-2011	08-01-2012
6.055	High	01-01-2012	05-01-2012
06.14	Medium	01-01-2012	06-01-2012
6.036	Medium	01-01-2012	01-08-2012
71.46	High	02-01-2012	11-01-2012
.1092	High	02-01-2012	11-01-2012
3.12	Medium	02-01-2012	06-01-2012
.8005	High	03-01-2012	07-01-2012
50.13	Medium	03-01-2012	09-01-2012
9.464	Low	03-01-2012	09-01-2012
7.472	Medium	03-01-2012	07-03-2012
.9018	Medium	03-01-2012	08-01-2012
0.412	Critical	03-01-2012	03-01-2012
7.77	High	03-01-2012	08-01-2012
42.9	Medium	03-01-2012	07-01-2012
53.32	High	03-01-2012	12-01-2012
90.43	Medium	03-01-2012	07-01-2012
26.76	Medium	03-01-2012	12-01-2012
7.856	Medium	04-01-2012	11-01-2012
09.84	Medium	04-01-2012	09-01-2012
null	High	04-01-2012	09-01-2012
-21.9	Medium	04-01-2012	09-01-2012
3.6	Medium	04-01-2012	11-01-2012
.5512	High	04-01-2012	08-01-2012
13.68	Medium	04-01-2012	08-01-2012
.3924	Medium	04-01-2012	04-04-2012
22.92	Low	05-01-2012	11-01-2012
99.36	null	05-01-2012	09-01-2012
20.52	Medium	05-01-2012	07-01-2012
4.65	Medium	05-01-2012	14-01-2012
.7748	High	05-01-2012	09-01-2012
35.64	High#	06-01-2012	09-01-2012
.0084	High	06-01-2012	10-01-2012
4.884	Low	06-01-2012	13-01-2012
13.11	null	06-01-2012	10-01-2012

Table.TransformColumnTypes(#"Replaced Value19",({{"Order Date", type date}, {"Ship Date", type date}}))

Order ID	Order Date	Ship Date	Customer Name	Country	Region	Category	Sub-Category
1	AO-2012-H097804-41105	15-07-2012	Jose O'Brian	Angola	Central Africa	Office Supplies	Paper
2	AO-2012-SW104554-40942	09-02-2012	Hauin Weien	Angola	Central Africa	Office Supplies	Paste
3	AO-2013-CC24754-41612	04-12-2013	Andy Chapman	Angola	Central Africa	Office Supplies	Art
4	AO-2014-HG48454-41863	12-08-2014	Barry Greene	Angola	Central Africa	Office Supplies	Storage
5	AU-2015-PF91208-42181	26-06-2015	Peter Fuller	Austria	Western Europe	Furniture	Art
6	CA-2012-AD10180140-41231	18-11-2012	Man Dominguez	United States	Eastern US	Office Supplies	Phone
7	CA-2012-AJ10960140-41165	13-09-2012	Astrea Jones	United States	Eastern US	Office Supplies	Books
8	CA-2012-AJ10555140-41004	05-04-2012	11-04-2012	United States	Western US	Office Supplies	Books
9	CA-2012-BF11020140-40955	16-02-2012	20-02-2012	United States	Western US	Office Supplies	Copies
10	CA-2012-CA12775140-41087	27-06-2012	01-07-2012	United States	Eastern US	Office Supplies	Supplies
11	CA-2012-CK12325140-41061	01-06-2012	06-06-2012	Christine Kargatis	United States	Western US	Art
12	CA-2012-OP13165140-41244	01-12-2012	03-12-2012	David Philippe	United States	Eastern US	Chair
13	CA-2012-OP13140140-41192	10-10-2012	15-10-2012	Sean Percer	United States	Western US	Technology
14	CA-2012-ES14080140-41183	01-10-2012	04-10-2012	Erin Smith	United States	Western US	Office Supplies
15	CA-2012-GK14620140-41084	24-06-2012	26-06-2012	Grace Kelly	United States	Western US	Office Supplies
16	CA-2012-GM14680140-41131	10-08-2012	14-08-2012	Greg Matthias	United States	Western US	Office Supplies
17	CA-2012-GM14695140-41003	04-04-2012	08-04-2012	Greg Maxwell	United States	Southern US	Office Supplies
18	CA-2012-HM14860140-41110	20-07-2012	24-07-2012	Barry Marie	United States	Central US	Furniture
19	CA-2012-JL15835140-41170	18-09-2012	23-09-2012	John Lee	United States	Southern US	Office Supplies
20	CA-2012-KM16375140-41273	30-12-2012	06-01-2013	Catherine Murray	United States	Eastern US	Furniture
21	CA-2012-MC17590140-41048	19-05-2012	21-05-2012	Matt Collister	United States	Central US	Technology
22	CA-2012-MY18295140-41026	27-04-2012	05-02-2012	Muhammed Yedwab	United States	Eastern US	Office Supplies
23	CA-2012-MY18295140-41236	23-11-2012	23-11-2012	Muhammed Yedwab	United States	Eastern US	Technology
24	CA-2012-NC18535140-41140	19-08-2012	24-08-2012	Nick Crebassa	United States	Southern US	Office Supplies
25	CA-2012-NW18400140-41080	20-06-2012	24-06-2012	Katalie Webber	United States	Eastern US	Office Supplies
26	CA-2012-RB19435140-41236	29-11-2012	28-11-2012	11-11-2012	United States	Central US	Technology
27	CA-2012-RP19390140-41052	29-05-2012	29-05-2012	Leslie Pflinking	United States	Eastern US	Office Supplies
28	CA-2012-SC20020140-41222	09-11-2012	15-11-2012	Sam Craven	United States	Central US	Office Supplies
29	CA-2012-SM20950140-41213	31-10-2012	07-11-2012	Luzanne McInair	United States	Central US	Office Supplies
30	CA-2013-AG10900140-41522	05-09-2013	09-09-2013	11-09-2013	United States	Western US	Office Supplies
31	CA-2013-AZ10750140-41406	12-05-2013	16-05-2013	Annlie Zypern	United States	Eastern US	Office Supplies
32	CA-2013-OP13390140-41404	10-05-2013	10-05-2013	Dennis Pardue	United States	Central US	Office Supplies
33	CA-2013-EA14035140-41604	26-11-2013	01-12-2013	Erin Ashbrook	United States	Central US	Technology
34	CA-2013-LI15850140-41423	29-05-2013	03-06-2013	John Lucas	United States	Western US	Office Supplies
35	CA-2013-LS1730140-41441	16-06-2013	20-06-2013	Lycoris Saunders	United States	Central US	Furniture
36	CA-2013-MB17305140-41541	24-09-2013	24-09-2013	Maria Bertelson	United States	Eastern US	Office Supplies

Query Settings

PROPERTIES

Name

orders

All Properties

APPLIED STEPS

Cleaned Text1

Split Column by Delimiter

Removed Columns

Renamed Columns

Split Column by Delimiter1

Removed Columns1

Renamed Columns1

Replaced Value2

Replaced Value

Replaced Value1

Removed Top Rows

Changed Type with Locale

Replaced Errors

Removed Top Rows1

Changed Type1

Replaced Value3

Replaced Value4

Replaced Value5

Replaced Value6

Replaced Value7

Replaced Value8

Replaced Value9

Replaced Value10

Replaced Value11

Replaced Value12

Replaced Value13

Replaced Value14

Replaced Value15

Replaced Value16

Replaced Value17

Replaced Value18

Replaced Value19

Changed Type2

Summary

- Removed duplicate records based on **Order ID**.
- Detected and handled missing (null) values in both text and numeric columns.
- Standardized column names and assigned appropriate data types.
- Cleaned **Order Date** and **Ship Date** columns by correcting date formats.
- Applied **Proper Case**, **Trim**, and **Clean** to text columns such as Customer Name, Country, Region, and Category.
- Replaced null values in text columns with **"Unknown"** using conditional logic.
- Replaced errors and missing values in numerical columns using **Average** and **Median** values.
- Corrected spelling inconsistencies and standardized text entries.
- The dataset was successfully prepared for data modelling and reporting.

Observations

- Duplicate Order IDs were removed, ensuring each transaction is unique.
- Missing values in text columns were replaced with **"Unknown"** to avoid blank records.
- Customer Name, Country, Region, and Category columns now have consistent formatting using Proper Case.
- Extra spaces and unwanted characters were removed using **Trim** and **Clean** functions.
- Order Date and Ship Date were standardized for accurate time-based analysis.
- Numerical columns no longer contain errors or missing values after applying Average and Median replacements.
- Spelling inconsistencies were corrected, improving data consistency.

8. The cleaned dataset is reliable, consistent, and ready for building relationships, creating DAX measures, and developing dashboards in Power BI.